

NULLQUAKE

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AGENDA

1. Problem statement
2. Feature spaces
3. Introduction to Machine Learning
4. Model selection
5. Training methodology
6. Results
7. Summary & Conclusion

⚠ DISCLAIMER ⚠

- Group Dynamics:
 - Math
 - CS/Math
 - Physics
 - Industrial Engineering
- None of us had prior ML experience

OBJECTIVE 1: Learn

OBJECTIVE 2: Have
FUN!!

PRESENTATION PHILOSOPHY:

A beginner-friendly approach to ML using quantum applications

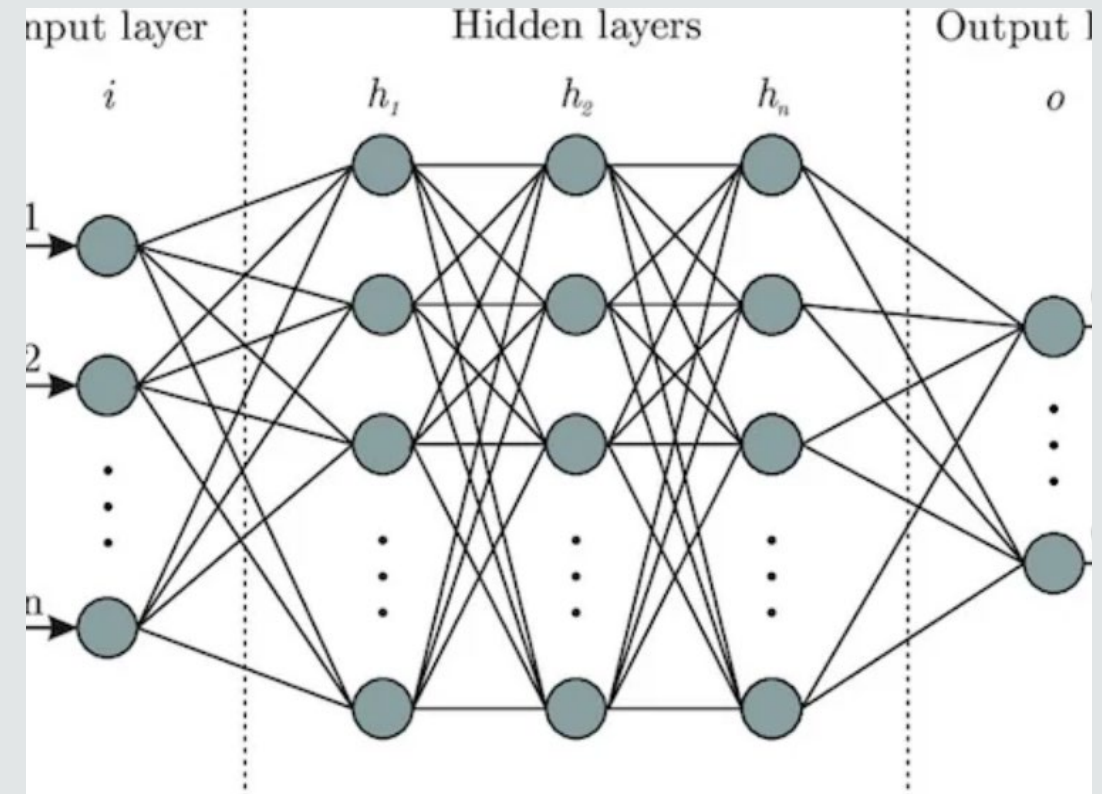
PROBLEM STATEMENT

1. Predict the runtime of a quantum circuit in the QuantumRings simulator.
2. 532 circuits, yielding 1596 different data points for training.
3. Each datapoint had a qasm file, a threshold, and the total runtime.
4. Do it all without spending more than 15 seconds on an output.

WHAT IS MACHINE LEARNING?

- Pretraining
- Lots and lots of linear algebra.
- These vectors span what we call a feature space
- Gradient descent

NEURAL NETWORK DIAGRAM



FEATURE SPACES

- What is a feature space?
- What makes a good feature space?
- The team disagreed internally:
 - Maximalist approach: Extract all possible information from a quantum circuit. Allow a model to work with all parameters
 - Minimalist approach: Reduce the feature space to a common gateset to minimize training costs
- To resolve the issue: We looked at our problem statement.

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4. **Do it all without spending more than 15 seconds on an output.**

Maximal approach won

MAXIMAL FEATURE SPACE

- 94 variables
- File name
- # qubits
 - Active
 - Inactive
- # operations
 - 1-qubits
 - 2-qubit
 - 2-qubit density
 - Multi-qubit
 - Entangling gate density
 - Parametrized
- Depth
- Parallelism
- Max gate parity
- Average qubit load
- Max qubit load
- Unique interactions
- Graph density
- Connected components
- Tree width
- Average 2-qubit distance
- Max 2-qubit distance
- Max cut width
- Max weighted degree
- Custom circuits

MAXIMAL FEATURE SPACE 2

Parametrized

Rotation

Clifford

X

Y

Z

H

T*****

S

U1

U2

U3

Sx

Rx

Ry

Rz

Rxx

Ryy

Rzz

MCX

MCZ

SWAP

XX-YY

$$XX + YY$$

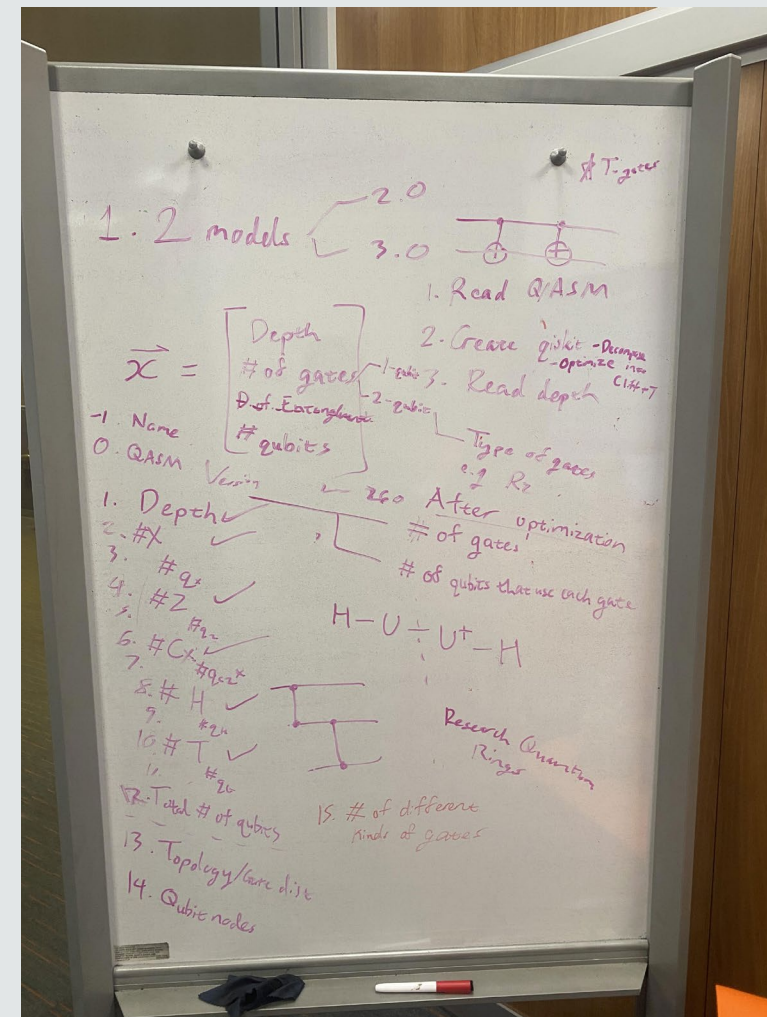
iSWAP

QFT

Controlled versions

Inverses

Custom gates



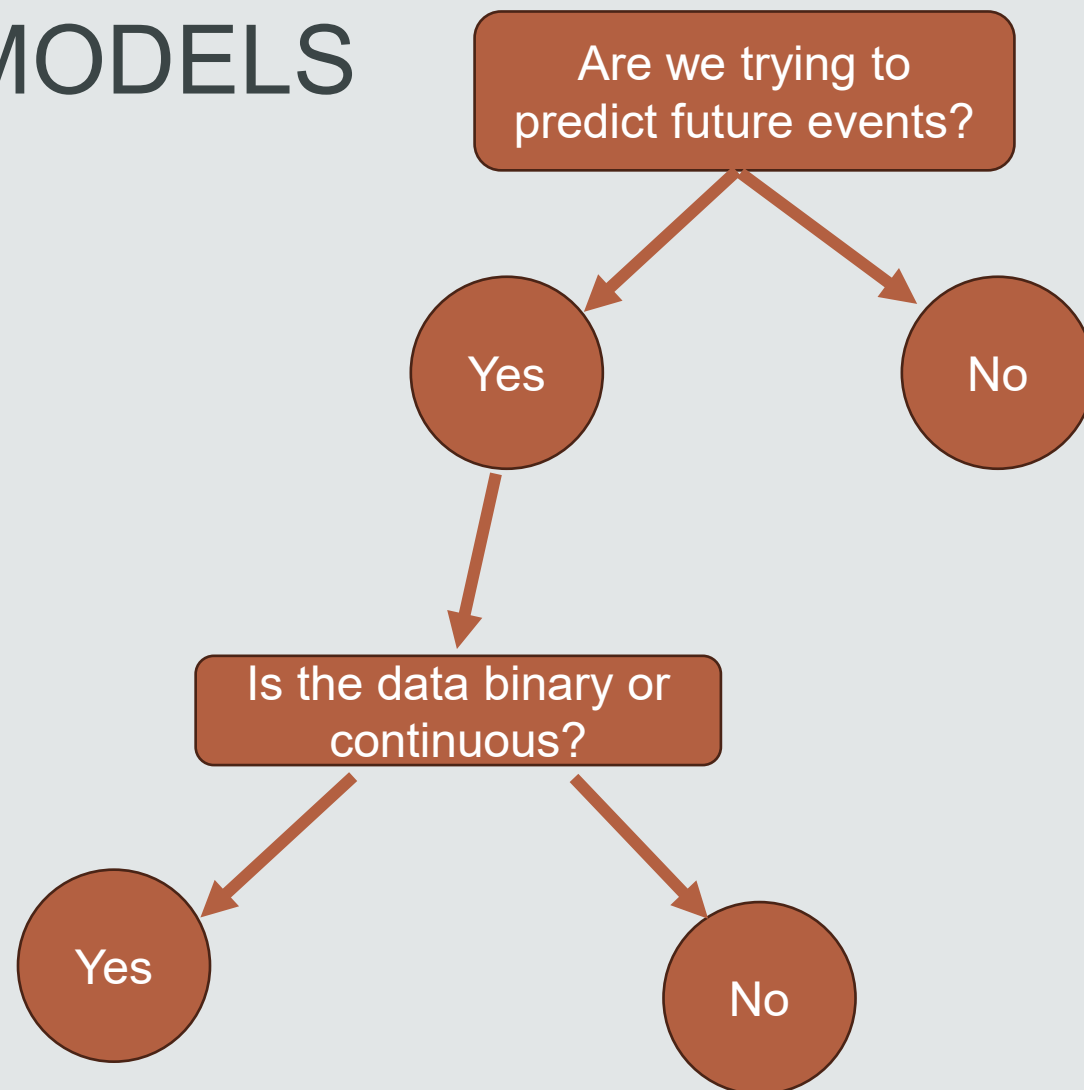
CHALLENGES: READING THE FILES

- The uncompressed files crashed 2 branches and forced github migrations
- These were later resolved
- Size of files was also hard to parse

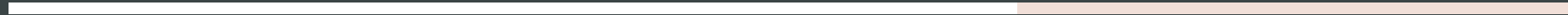


MACHINE LEARNING MODELS

- Main continuous-output models include:
- Linear regression
- K-Nearest Neighbour (KNN)
- Gradient-boosted trees
- Random forests
- Support Vector Machines
- XG-Boost
- Combined ensemble methods and more!

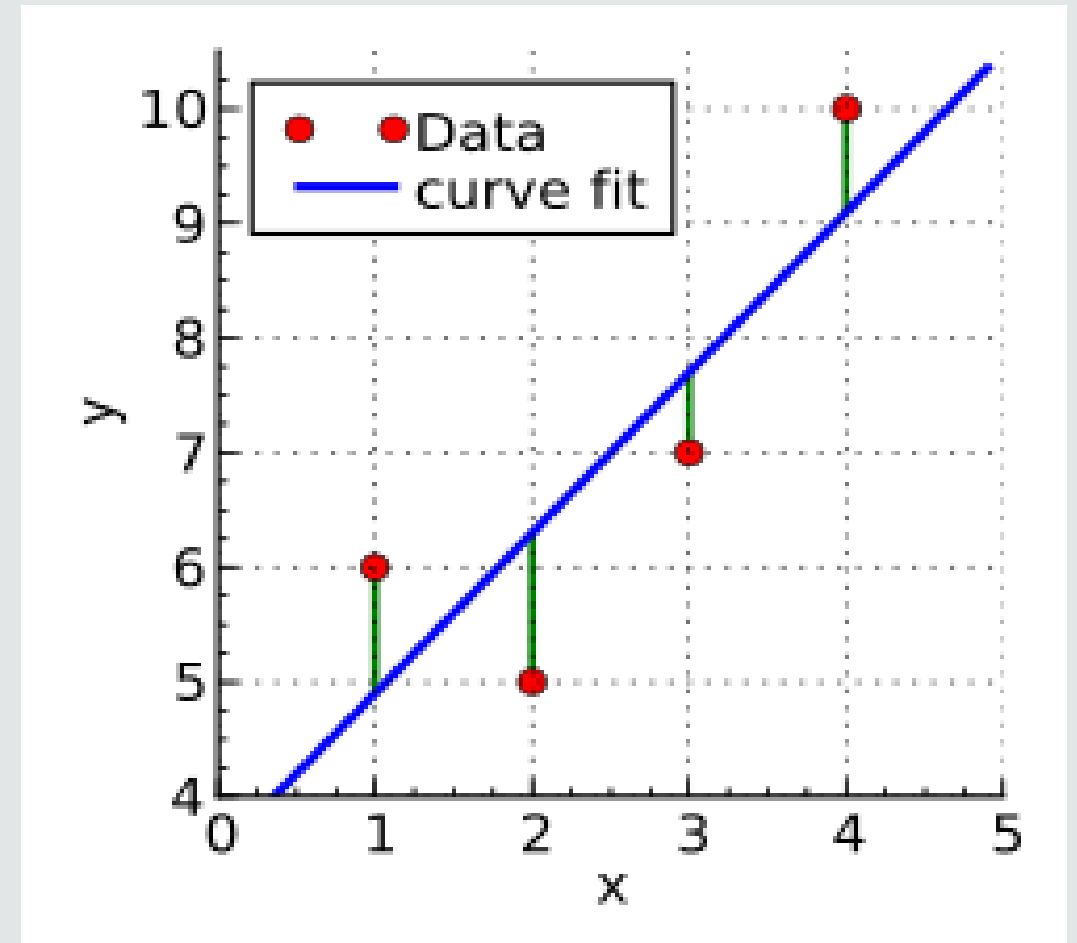


MODEL BLITZ



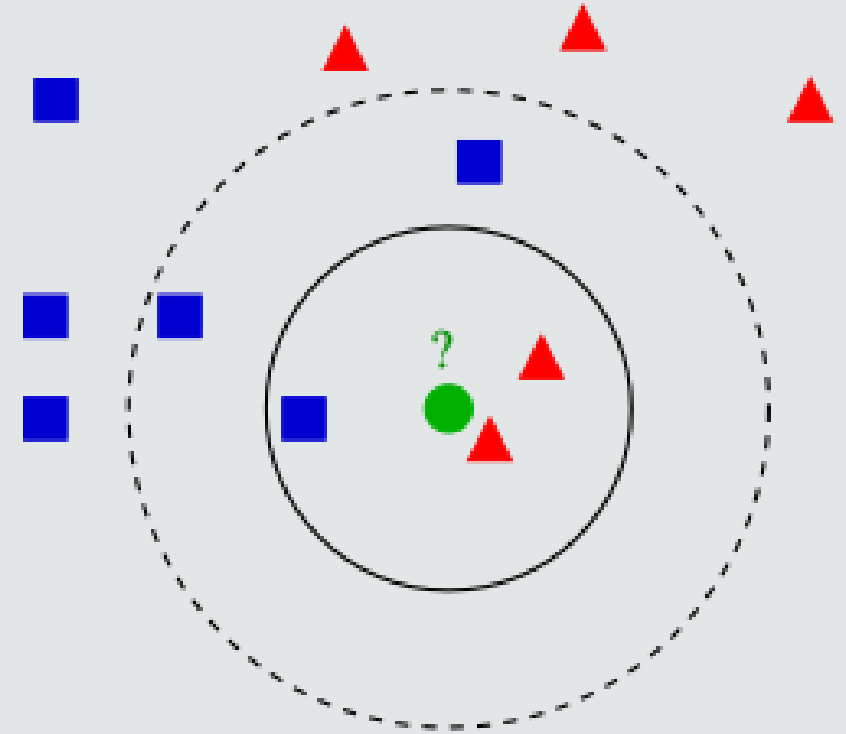
LINEAR REGRESSION

- The model fits a best-fit hyperplane to the data



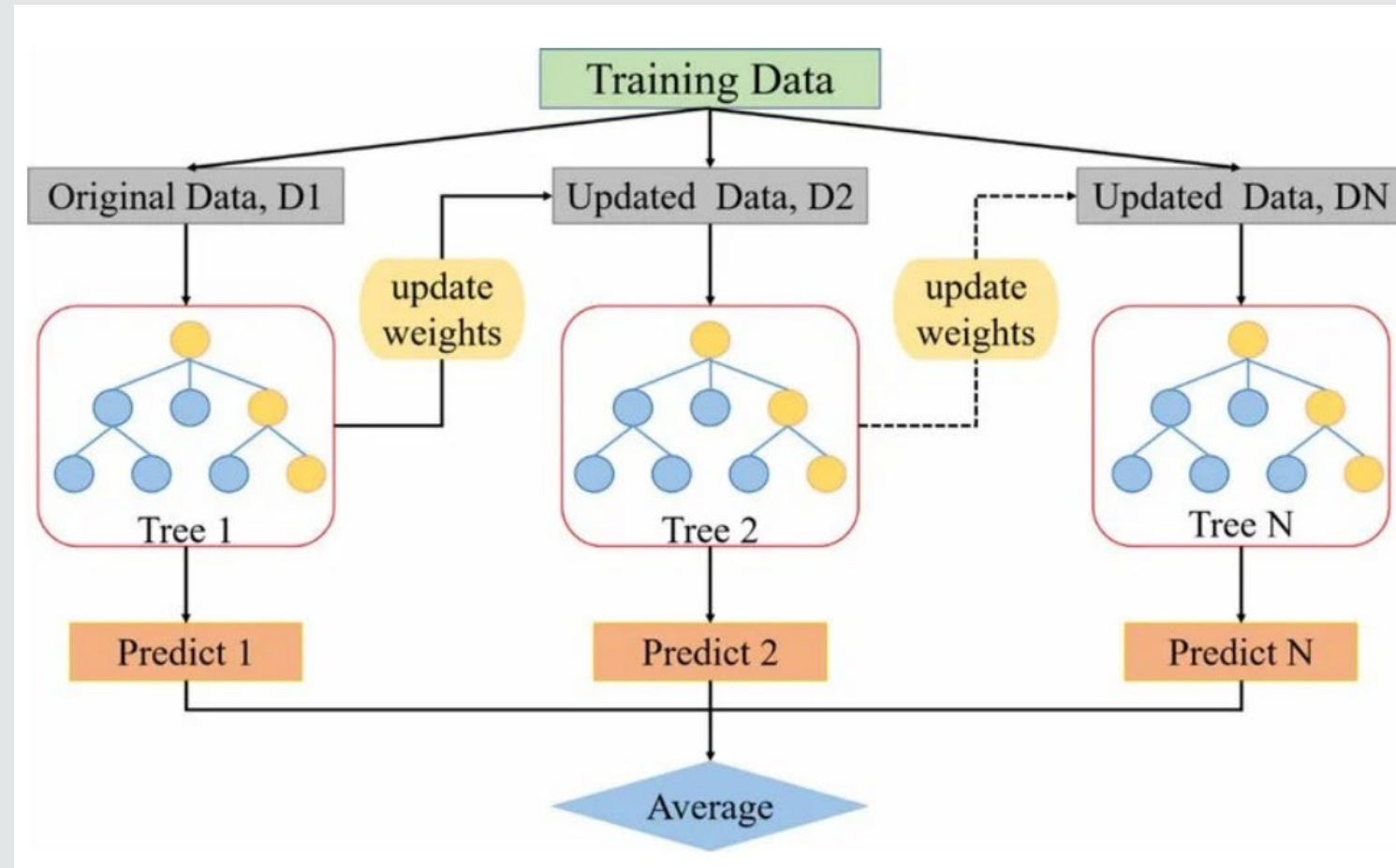
K-NEAREST NEIGHBOUR (KNN)

- Averages the K nearest neighbors



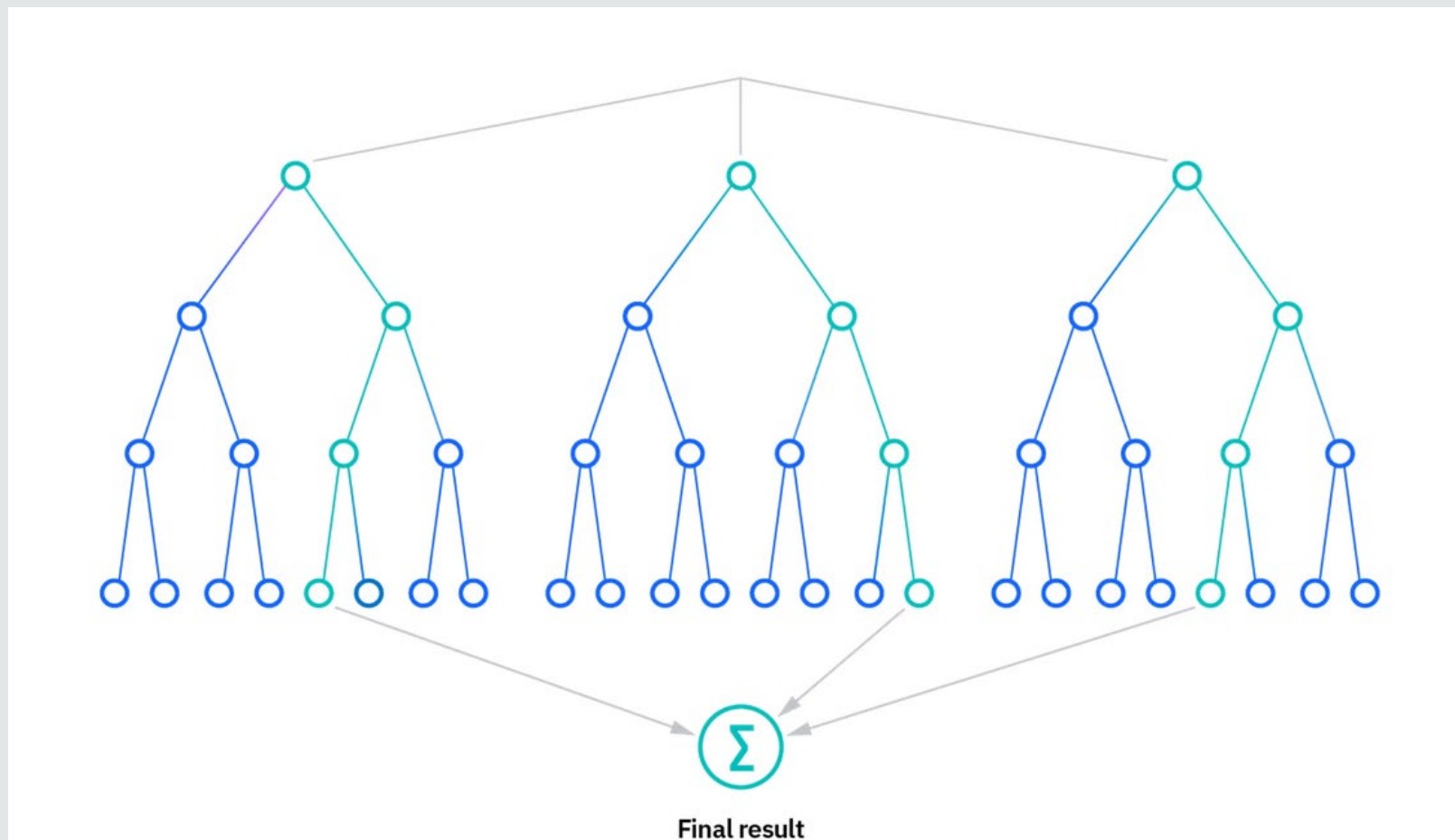
GRADIENT-BOOSTED TREES

- A series of decision trees where each tree uses previous results to train.



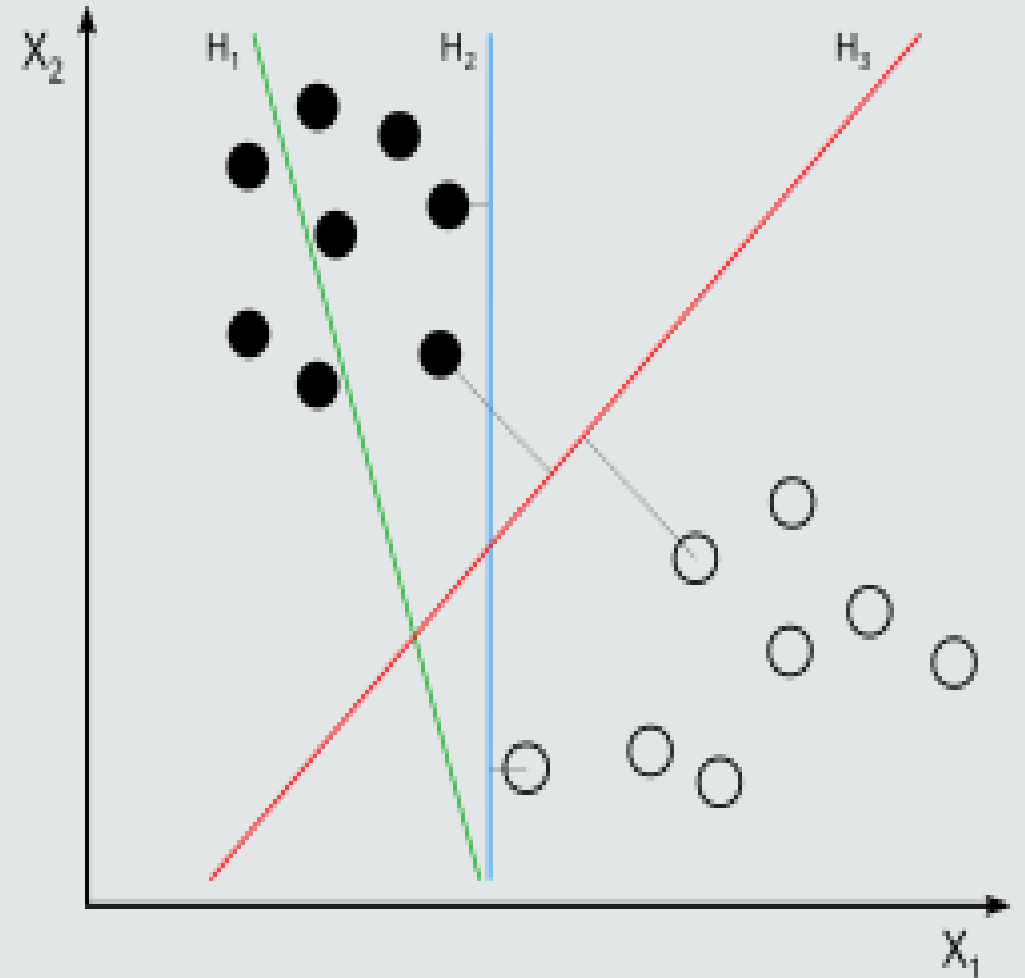
RANDOM FORESTS

- A population of trees
- Each trained on a random subset of the data
- Outputs averaged.



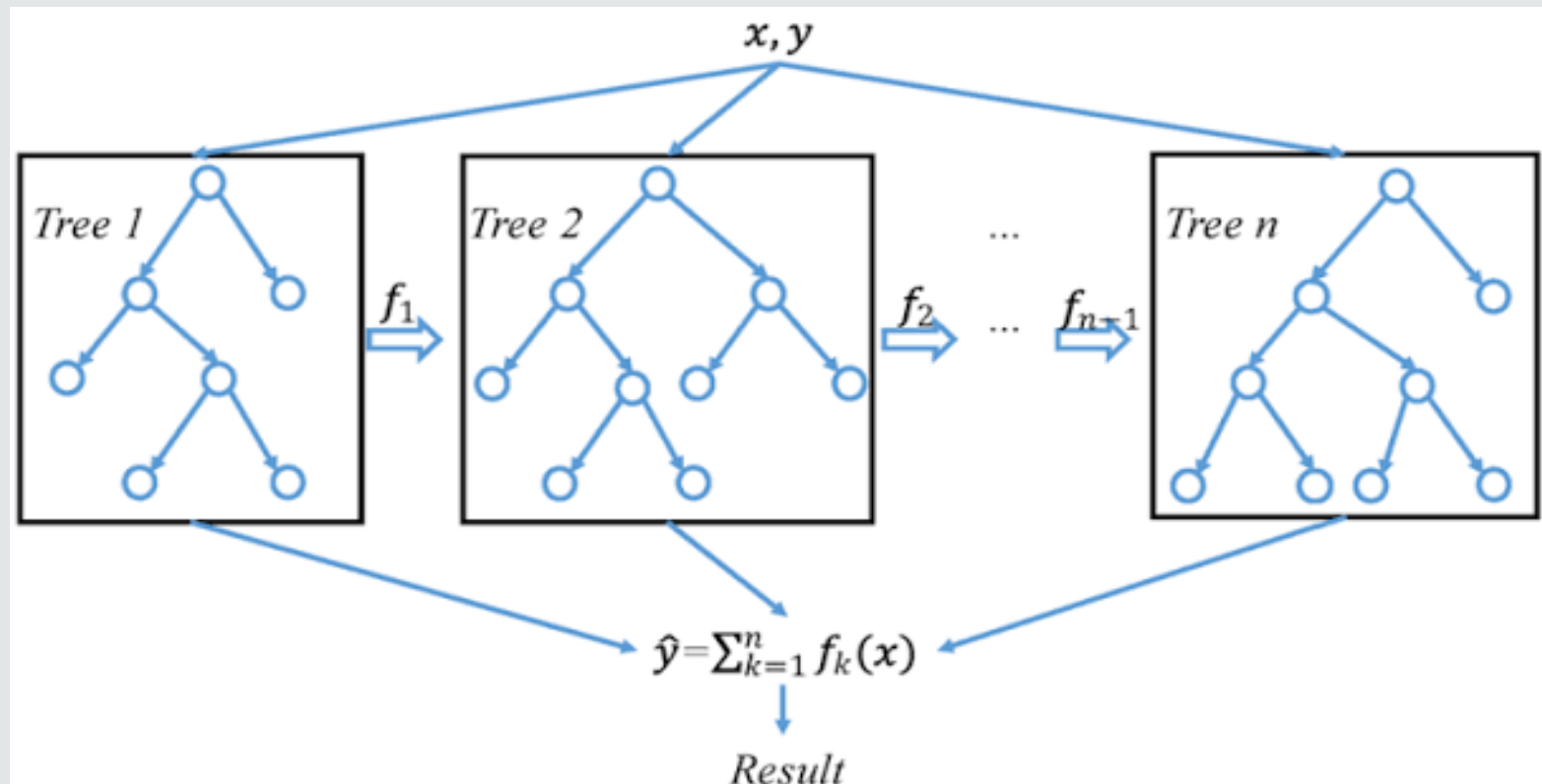
SUPPORT VECTOR MACHINES

- Finds hyperplanes to divide the datapoints into different categories.



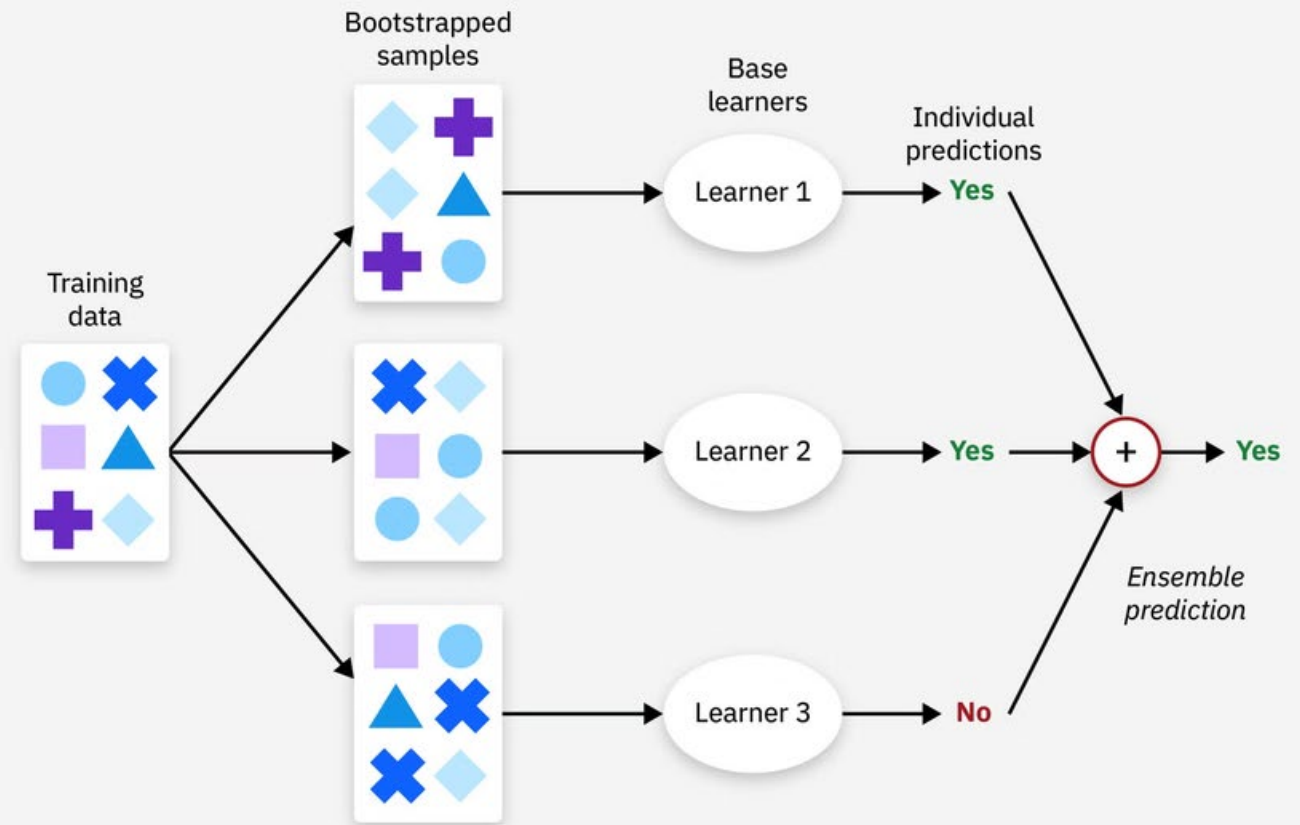
XG-BOOST

- Library for gradient boosting
- Basically gradient boosted trees.



COMBINED ENSEMBLE METHODS

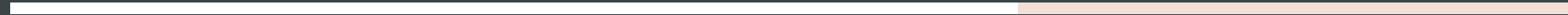
- Method of combining multiple models.



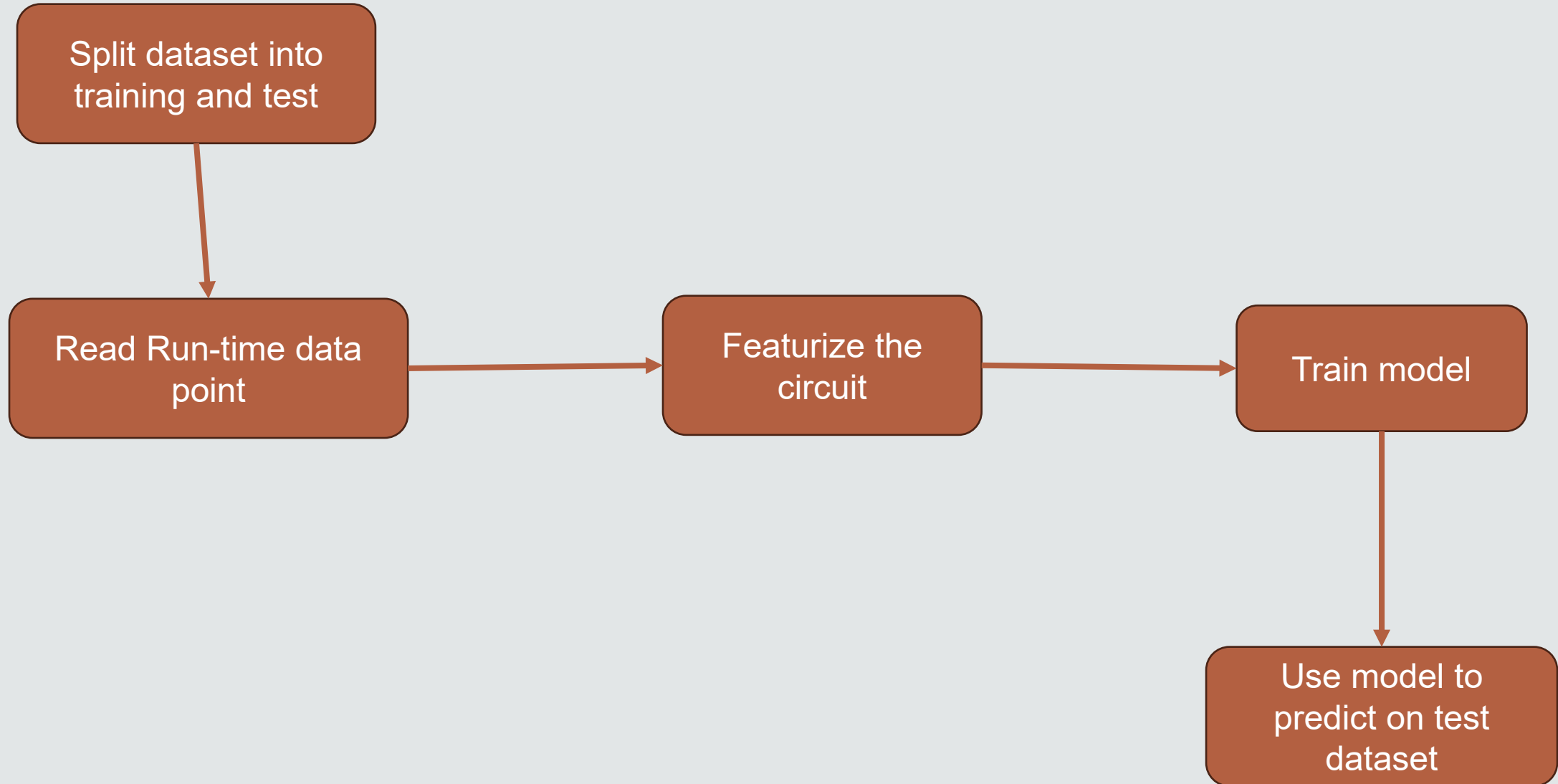
OUR THEORETICAL FINALISTS

Model	Pros / Justifications	Cons
K-Nearest Neighbor	Straightforward logic	Which neighbors count as close?
Random Forest	Multiple trees like quantum randomness	slow prediction times, large memory and storage footprints, lack of interpretability
Combined Ensemble	Best of different models	Worst of different models
XG - Boost	Learns non-linear interaction well	poor extrapolation, timeout examples, loss of circuit structure

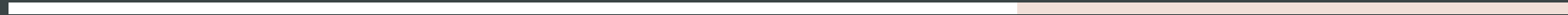
FINAL WORKFLOW



TRAINING PIPELINE

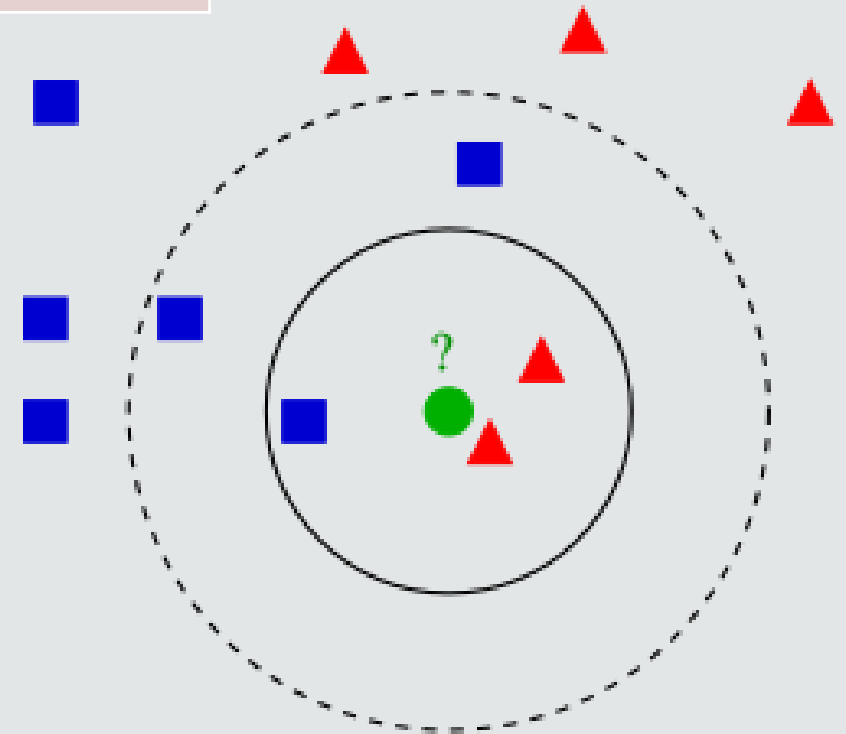


RESULTS



WINNER: K-NEAREST NEIGHBOUR!!

Model	Average accuracy	16	64	512
K-Nearest Neighbor	96	96	96	95

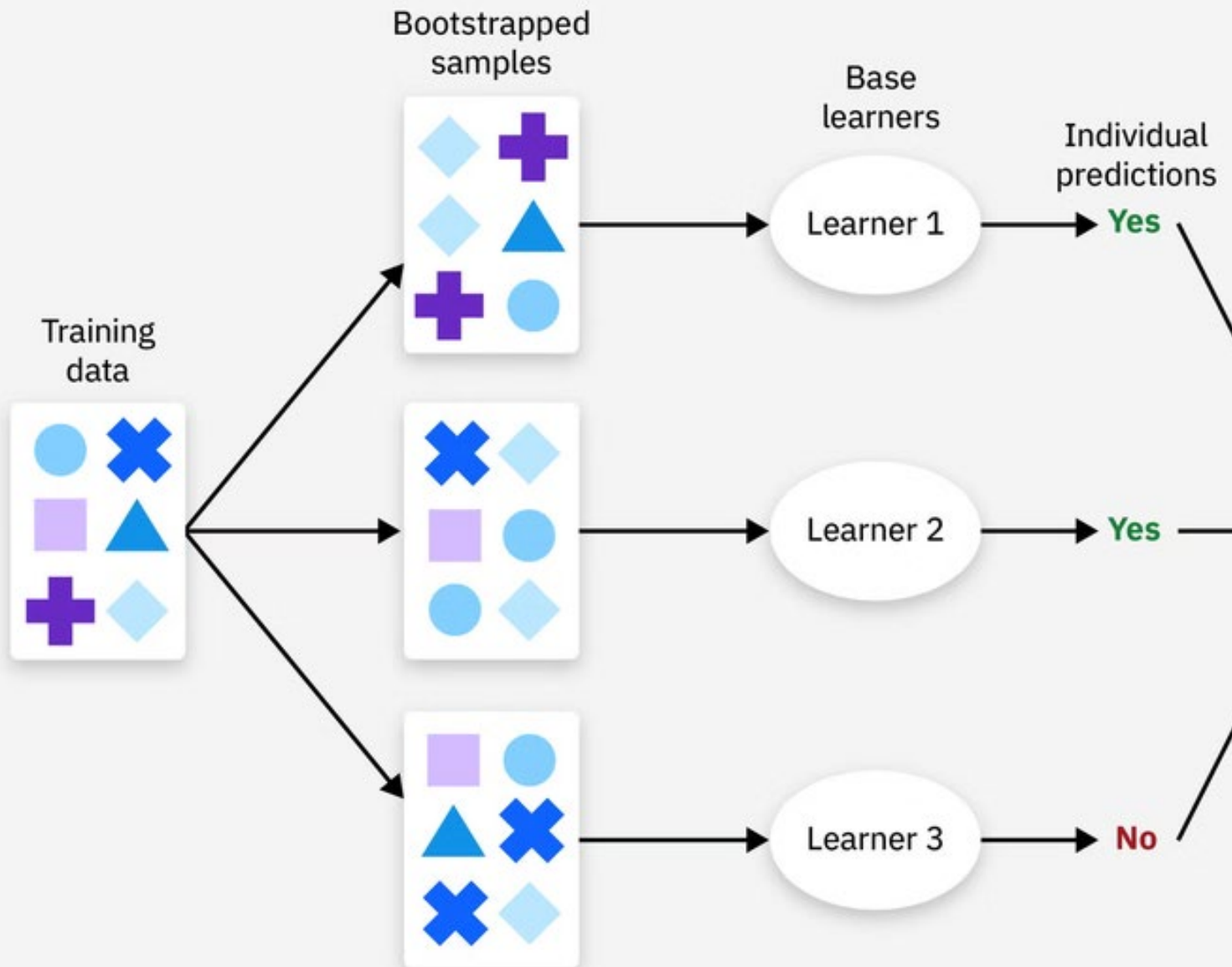


THAT WASN'T TRUE...

Challenge 2: Five-fold reduction

OUR FINALISTS

Model	Average accuracy	16	64	512
K-Nearest Neighbor	86	86	86	84
Random Forest	87	89	87	85
Combined Ensemble: Extra Trees (70%), Light GBM (20%), Hist Gradient Boosting (10%)	89	91	89	88
XG - Boost	84	85	82	80



WINNER: COMBINED ENSEMBLE

Extra Trees: 70%

Light GBM: 20%

Hist Gradient Boosting: 10%

WHAT IF WE KEPT
GOING?



HOW MUCH CAN
WE REDUCE THE
FEATURE SPACE
WITHOUT
DEGRADING
PERFORMANCE?

FEATURES VS. AVERAGE ACCURACY

Non-populated features removed

Features	Accuracy (%)
94+1	89.21
61+1	89.11
3+1	9

FUTURE WORK

CORRECTLY
DETERMINING THE
TRADEOFF BETWEEN
FEATURE SPACE
ACCURACY AND SIZE

TAKING ADVANTAGE
OF CIRCUIT
STRUCTURES
(COMMUTING GATES)

OTHER MACHINE
LEARNING MODELS /
COMBINATIONS OF
COMBINED
ENSEMBLE

THANK YOU!!

Questions?
